# *Quiz 9 Reflection*


SCIE 001 Physics

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## Contents

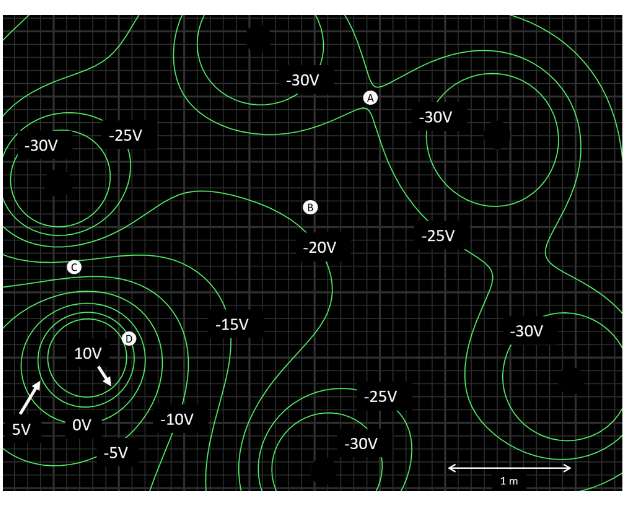
# 1. A Potential Landscape

## 1.1. Specify Question

Question 6: A potential landscape

The following diagram shows several equipotentials due to some source charges, which are not shown. Consider the labeled points (i.e. the small white circles) and complete the following sentence:

▸ Hint #1




"Point B ✕ 0% has zero electric field, while point D ✓ 100% has the largest magnitude of electric field of the four points."

Correct answer

"Point A has zero electric field, while point D has the largest magnitude of electric field of the four points."

## 1.2. Diagnosis Phase

I knew that electric field is equal to the gradient of the electric potential.

However, I incorrectly thought that this results in the greatest magnitude of electric field being where the equipotential lines were closest, and the least magnitude of electric field being where they were furthest apart.

I forgot to consider the change in potential of the equipotential lines, causing me to overlook the saddle point when reading the graph (seeing the equipotential lines so close together), and instead chose the next best option with far equipotential lines (B).

Words: 90

## 1.3. Correction Phase

Looking at the graph correctly we see:

Point D has the greatest slope, and therefore the greatest electric field.

Point C is not a saddle point, because it is between two charges of opposite signs (one positive, one negative), and therefore has some electric field.

Point B, while having a small magnitude, has some electric field.

And finally point A is a saddle point between two charges, having 0 electric field.

Therefore Point A has zero electric field, and point D has the largest magnitude of electric field.

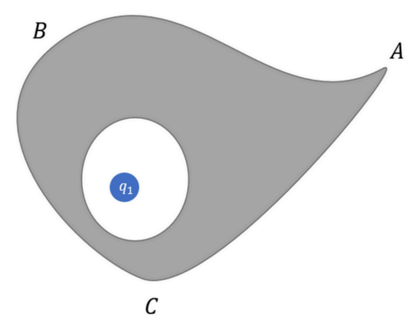
Words: 88

# 2. A Conductor with a Cavity

## 2.1. Specify Question

Question 9: A conductor with a cavity

As shown below, a point charge $q_1 = 10\,\mathrm{nC}$ is fixed in an otherwise empty cavity that was carved out of a solid conducting object.




a) If the conducting object has a net charge of $Q = -20\,\mathrm{nC}$, how much charge is on its outer surface?

$Q_{\text{outer}} =$ 0 nC ? ✕ 0%

b) How does the magnitude of the surface charge density rank at the various points labeled above?

- $\sigma_C > \sigma_B > \sigma_A$
- $\sigma_C < \sigma_B < \sigma_A$
- None of the other options.
- $\sigma_A = \sigma_B = \sigma_C$
- $\sigma_A > \sigma_C > \sigma_B$ ✓
- $\sigma_A < \sigma_C < \sigma_B$

✓ 100%

Correct answer

$Q_{\text{outer}} =$ `-10 nanocoulomb`

$\sigma_A > \sigma_C > \sigma_B$

## 2.2. Diagnosis Phase

I knew that because the object was a conductor, the charge it held was free to redistribute itself within the object. However I did not make the connection between this property and gauss's law, causing me to become confused when the object was charged.

I knew that if $q_1$ was 10 nC and the object was neutral, the inner surface would have −10nC of charge, and the outer surface would carry 10 nC of charge. $-10\text{ nC} + 10\text{ nC} = 0\text{ nC}$

I guessed 0 nC because I similarly pictured all the −20 nC of charge rushing towards the positive $q_1$, leaving no charge on the outer surface. $-20\text{ nC} + 0\text{ nC} = -20\text{ nC}$

Words: 99

## 2.3. Correction Phase

To solve this, we can be smarter about the properties of conductors. Since we know they are free charge carriers, in an electric field, their charges must rearrange, generating an equal and opposite opposing field.

Therefore conductors (at equilibrium) must have no internal electric field, and hence constant electric potential.

Imagine a gaussian surface within the object around its centre, since $E = 0$, flux is zero, therefore all charge enclosed must add to 0. Therefore its inner surface must have:

$$\begin{aligned} q_{\text{Particle}} + q_{\text{Inner Surface}} &= 0 \\ q_{\text{Inner Surface}} &= -q_{\text{Particle}} \\ q_{\text{Inner Surface}} &= -10\text{ nC} \end{aligned}$$

But, the charge of the object must be −20 nC, therefore its inner and out surface must sum to this value.

$$\begin{aligned} q_{\text{Inner Surface}} + q_{\text{Outer Surface}} &= -20\text{ nC} \\ q_{\text{Outer Surface}} &= -20\text{ nC} - q_{\text{Inner Surface}} \\ q_{\text{Outer Surface}} &= -20\text{ nC} - (-10\text{ nC}) \end{aligned}$$

$$\therefore q_{\text{Outer Surface}} = -10\text{ nC}$$

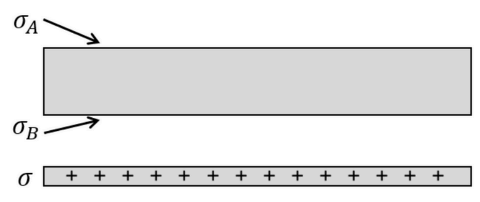
Words: 99

# 3. Comparing Surface Charges

## 3.1. Specify Question

Question 10: Comparing surface charges

As shown below, a large conducting plate of some finite thickness is held above and parallel to a thin plane of charge with surface density $\sigma$. You can assume that both objects are infinite in extent. Rank the magnitude of the surface charge densities.




- $\sigma = \sigma_A = \sigma_B$
- $\sigma > \sigma_A > \sigma_B$
- $\sigma < \sigma_A < \sigma_B$
- $\sigma = \sigma_A > \sigma_B$ ✕
- $\sigma > \sigma_A = \sigma_B$

✕ 0%

Correct answer

$\sigma > \sigma_A = \sigma_B$

## 3.2. Diagnosis Phase

For this question I made a mistake in reading the question. I thought that it was asking me to rank the surface charge densities, not the magnitude of the surface charge densities.

I also forgot to account for the electric field dissipating over distance.

These mistakes caused me to see the line of positive charge causing the charges within the conducting plate to rearrange, where the charge density $\sigma_B$ is equal to $\sigma$ but negative, and $\sigma_A$ equal but positive.

If the plates were up against each other (no dissipation of electric field) and using signs:

$$\sigma_B = -\sigma \qquad \sigma_A = \sigma$$

$$\text{So} \quad \sigma_B < \sigma = \sigma_A$$

$$\therefore \sigma = \sigma_A > \sigma_B$$

Words: 93

## 3.3. Correction Phase

The question was asking for the "magnitude" of surface charge density. And the electric field for an infinite line of charge is related to $\sim \frac{1}{r}$ where $r$ is the distance to the line.

When exposed to an electric field, the charges in a conductor move to the surfaces, generating an opposing field that perfectly cancels it out.

The line of charges generate an electric field, causing charges to redistribute in the conducting plate.

Since the conductor is neutral:

$$|\sigma_A| = |\sigma_B|$$

And since the electric field reduces with distance from the line of charge:

$$|\sigma| > |\sigma_A|$$

$$\therefore |\sigma| > |\sigma_A| = |\sigma_B|$$

Words: 89